SATYA-APAAR 2.1

Enhanced Final Proposed Solution

Student Admission Trust & Yield Assurance for AICTE-approved institutions

Core positioning
One student. One academic identity. Every admission traceable. Every approved seat accounted for.

Problem Statement: SIH26263 — APAAR ID Based Student Database Management & Verification Portal

Theme: Smart Automation | Software

Document status: Enhanced design proposal — not an official AICTE specification

Version: 2.1 | **Date:** 30 October 2026

This revision preserves the original SATYA-APAAR 2.0 technical design and integrates improvements identified through comparison with another proposed approach for the same problem statement.

1. Executive Summary

AICTE receives student-related information from approved institutions through different submission mechanisms. Some submissions contain student-level details, while others provide aggregate enrolment counts. Differences in formats, identifiers, reporting periods and institutional systems make it difficult to build a complete, consistent and verifiable student registry.

SATYA-APAAR 2.1 is an AICTE-specific assurance layer that transforms institution-reported data into reconciled, time-aware and auditable enrolment records. It does not replace APAAR, ABC, DigiLocker or institutional ERP systems; it connects to authorised systems through adapters.

Solution in one line
SATYA-APAAR converts admissions into time-bounded enrolment episodes, reconciles student-level and aggregate reporting, checks each episode against a policy-versioned seat ledger, and routes unexplained cases to authorised human reviewers with evidence.

What is strengthened in version 2.1?

- **Aggregate-to-student-level reconciliation:** compares institution-reported totals with the count and status of submitted student-level records, while preserving unresolved historical or incomplete data as explicit exceptions.
- **Institution onboarding and submission journey:** adds registration, institution/programme context, academic-year selection, upload tracking, correction and resubmission states.
- **AICTE operational dashboard:** defines the actual metrics, filters, queues and drill-downs needed by officers.
- **Implementation tiers:** separates a demonstrable core prototype from advanced features and longer-term integrations.
- **Integration resilience:** distinguishes local format validation from genuine APAAR registry verification and handles unavailable services without mislabelling records as fraudulent.

Design principles

- A matched identity is not automatically a duplicate admission.
- A duplicate candidate or anomaly is not a fraud verdict.
- Aggregate totals and student-level counts must remain distinguishable.
- Rules must be derived from authorised AICTE approval evidence and applicable policy versions.
- Every consequential flag must have a reason, evidence reference, rule/model version and human review path.
- The prototype must clearly label simulated integrations and synthetic test data.

2. Problem Description and Gaps

2.1 Existing problem

- Fragmented student data submission across institutions and different formats.
- Limited student-level enrolment records where only aggregate totals are reported.
- Missing, inconsistent or non-standard student identifiers.
- Difficulty identifying the same student across institutions.
- Difficulty reconciling admissions with approved intake and category-specific limits.
- Limited visibility into historical changes, transfers, withdrawals and corrections.
- Privacy, interoperability and authorised verification constraints.

2.2 Why a basic upload-and-duplicate portal is insufficient

A platform that only accepts spreadsheets, checks APAAR format, searches for repeated identifiers and compares a total against intake can identify some data-quality issues. It may still misclassify legitimate transfers, fail to explain historical aggregate discrepancies, or use the wrong seat-policy version. SATYA-APAAR therefore separates identity matching, enrolment lifecycle interpretation, seat-policy reconciliation and risk review.

2.3 Newly explicit gap: aggregate versus student-level records

An institution may report an aggregate total for a programme and academic year without supplying a complete row for every student. A later student-level upload may cover only part of that total. The system must not silently treat the two values as equivalent.

Situation	System interpretation	Required action
Aggregate total = student-level active count	Counts reconcile for the defined scope	Mark reconciled and retain source references
Aggregate total is higher than student-level count	Possible missing student-level records or scope mismatch	Create reconciliation variance; request explanation or additional records
Student-level count is higher than aggregate total	Possible late upload, reporting error or status mismatch	Flag for correction and re-run reconciliation
Aggregate-only historical submission	No student-level identity resolution is possible for those records	Store as aggregate-only evidence; do not invent student identities
Different academic-year or programme scope	Counts are not directly comparable	Require scope alignment before comparison

Important safeguard
An aggregate count cannot be converted into named student records without evidence. SATYA-APAAR records the limitation, tracks the variance and requests authorised reconciliation rather than fabricating identities.

3. Proposed Solution — SATYA-APAAR 2.1

3.1 Main operating concept

1. Institution is onboarded and its approved institution/programme context is selected.
2. Institution submits student-level admission events, aggregate totals, or both, using portal, CSV/Excel or authorised API.
3. Smart Ingest maps fields into a standard schema and reports row-level data-quality issues.
4. Aggregate Reconciliation compares reported totals with eligible student-level records using the same institution, programme, category and academic-year scope.
5. APAAR Identity Resolver verifies through an authorised adapter and creates protected matching references.
6. TEIE constructs enrolment episodes and classifies new admission, transfer, transition overlap, policy exception, unexplained overlap or duplicate data entry.
7. SeatGuard checks the episode against approved intake and the applicable policy snapshot.
8. Integrity Intelligence adds explainable risk indicators; it does not make an automatic fraud decision.
9. Clean records are committed; exceptions enter the AICTE review queue with evidence.
10. Reviewers approve, reject or return records for correction. Every correction creates a new version.
11. Dashboards and authorised verification views update from the reconciled records.

3.2 Two operating modes

Mode	Purpose	Boundary
Post-admission mode — core	Validate submitted admissions, reconcile records and flag conflicts or over-cap conditions near real time.	Detection and review; not prevention of an admission already completed.
Pre-admission mode — advanced	Reserve a seat before admission confirmation where an authorised counselling/admission workflow is integrated.	Cannot be claimed from file upload alone; requires workflow integration and transactional reservation.

4. Institution Onboarding and Submission Workflow

Version 2.1 makes the institution-facing journey explicit so that data submission is operationally usable, not merely a backend capability.

1. **Institution registration:** institution identity and authorised operator accounts are created or provisioned by the system administrator/AICTE process.
2. **Institution context:** operator selects institution, programme, academic year, approved intake and applicable seat categories. Approved intake is loaded from the independent AICTE Approved Intake and Policy Master, not trusted from the uploaded file.
3. **Submission type:** choose student-level admission events, aggregate enrolment report, or both.
4. **Upload and mapping:** upload CSV/Excel or use an authorised API. Smart Ingest suggests column mapping; the operator confirms uncertain mappings.
5. **Preflight validation:** required fields, dates, formats, duplicate rows and logical consistency are checked. Errors are returned with row number, reason code and correction guidance.
6. **Reconciliation preview:** show aggregate total, student-level count, excluded/pending records and variance before final submission.
7. **Submit and track:** assign a submission ID and show states such as Draft, Preflight Failed, Submitted, Processing, Partially Reconciled, Needs Correction, Under Review and Accepted.
8. **Correction and resubmission:** corrected data supersedes the previous version; the prior submission remains available in the audit history.

Submission state	Meaning
Draft	Operator has started but not submitted the batch.
Preflight Failed	Required fields or file-level checks failed; corrections are needed.
Submitted / Processing	Batch is accepted for processing; validation and reconciliation are running.
Partially Reconciled	Some records reconcile, while variances or aggregate-only portions remain unresolved.
Needs Correction	Institution must correct or explain specific findings.
Under Review	A case requires authorised AICTE review.
Accepted / Reconciled	All applicable checks passed or exceptions were formally resolved.

5. Core Modules

Module	Function	Output
Smart Ingest	Portal, CSV/Excel and REST upload; assisted fuzzy column mapping; preflight validation; idempotency keys.	Clean, normalised events and row-level findings.

Module	Function	Output
Aggregate Reconciliation Engine — new explicit workflow	Compares aggregate submissions with student-level counts by institution, programme, year, category and status; tracks variance and provenance.	Reconciliation status, variance amount, missing-detail queue and explanation request.
APAAR Identity Resolver	Authorised adapter; keyed HMAC tokens; exact protected matching; Fellegi-Sunter candidate matching when APAAR is missing.	Verified, provisional or failed identity; protected match candidate.
TEIE — Temporal Enrolment Integrity Engine	Builds time-bounded enrolment episodes and classifies timeline patterns.	Transfer classification, conflict case and current enrolment projection.
SeatGuard — Policy-Versioned Seat Ledger	Calculates permitted capacity from approval evidence and effective-dated rules; tracks occupancy by institution, programme, year and category.	Seat ledger, vacancy, alerts and policy trace.
Integrity Intelligence	Deterministic checks first; Isolation Forest, median/MAD peer comparison and temporal graph analysis as risk signals.	Explainable indicators and review-priority score.
RecordTrust	Versioned corrections and hash-chained tamper-evident audit log.	Complete record history and audit verification.
ConsentPass — Credential & Verification	Signed W3C Verifiable Credential enrolment attestation; purpose-limited QR/API verification, status and revocation.	Minimal-claim verification and access log.

6. Aggregate-to-Student-Level Reconciliation Engine

6.1 Objective

Provide a controlled bridge between aggregate reporting and student-level records. The engine detects mismatches, preserves the source of each count and prevents incomplete historical submissions from being presented as fully verified student registries.

6.2 Reconciliation dimensions

- Institution ID and campus where applicable.
- Programme/course and approved intake reference.
- Academic year/session and reporting cut-off date.
- Seat category, including applicable regular and approved special pools.
- Enrolment status: active, withdrawn, cancelled, transferred, pending or other configured state.
- Submission version and source: portal, file, API or approved legacy import.

6.3 Reconciliation calculation

Illustrative calculation
Reported aggregate enrolment = 500; eligible student-level active records = 472; unexplained variance = 28. The system records a variance of 28, checks scope/status differences, and requests missing records or an authorised explanation. It does not create 28 fictional student profiles.

A basic variance can be represented as:

Variance = Reported aggregate count – **Eligible student-level count**

This calculation is meaningful only after institution, programme, academic year, category, status and reporting cut-off are aligned. The system should separately display explained exclusions, pending cases and unresolved variance.

6.4 Reconciliation outcomes

Outcome	Meaning	Handling
Reconciled	Counts match for the defined scope.	Store result and evidence references.
Variance — detail missing	Aggregate exceeds student-level count.	Request missing student-level details or explanation.
Variance — count mismatch	Student-level eligible count exceeds aggregate.	Return for correction or authorised confirmation.
Scope mismatch	Different year, programme, category or cut-off.	Do not compare until scopes align.
Aggregate-only	Only a total is available.	Preserve as aggregate evidence; identity checks remain unavailable.
Partially reconciled	Some categories or periods reconcile; others do not.	Show reconciled and unresolved portions separately.

6.5 Data model additions

- **Aggregate_Submission:** institution, programme, year, category, reporting date, reported count, source and version.
- **Reconciliation_Run:** aggregate submission ID, student-level submission scope, rule version, computed count and status.
- **Reconciliation_Variance:** variance value, classification, explanation, evidence and resolution status.
- **Reconciliation_Resolution:** actor, action, timestamp, supporting evidence and superseding version.

7. TEIE — Temporal Enrolment Integrity Engine

TEIE separates identity matching from enrolment-conflict interpretation. A protected-token match is only a candidate; the timeline and applicable programme/study-mode policy determine whether a review case is needed.

Timeline pattern	Classification	Handling
First admission for identity	New enrolment	Commit after required checks.
Previous episode ended before new start	Sequential transfer	Verify withdrawal/transfer event; commit.
Short overlap with withdrawal evidence	Transition overlap	Accept with evidence reference if permitted.
Overlap allowed by configured policy	Policy-based exception	Check programme restrictions and log rule version.
Prolonged overlap without withdrawal or policy basis	Unexplained overlap	Create human review case.
Same institution/programme, multiple active records	Potential duplicate data entry	Request correction.
Same student repeated in one upload	Duplicate submission	Idempotency and row validation.

Key distinction
The same APAAR identity appearing in two institutions does not itself prove duplicate admission or fraud. TEIE checks effective dates, recorded dates, withdrawal evidence, study mode and configured policy.

8. SeatGuard — Policy-Versioned Seat Ledger

- Ledger key: institution, programme, academic year and seat category.
- Approved intake and category rules originate from an independent AICTE Approved Intake and Policy Master, not institution-submitted values.
- Supernumerary, NRI, OCI and foreign-national categories are maintained as separate pools with exact rule references and configured combined caps where applicable.
- Each decision stores handbook/notification identifier, effective-from and effective-to dates, programme applicability and parameters.
- Configurable alert thresholds are operational settings, not claims about statutory limits.
- Aggregate-versus-student-level variances are visible alongside seat utilisation.
- Database transactions, row locking and idempotency keys prevent concurrent requests from taking the same final seat.
- A seat is released only by an authorised withdrawal, cancellation, transfer or other verified status event — never because a later CSV omits a student.

Illustrative example
Approved intake 100; admitted 120; approved supernumerary allowance 10. Ten admissions remain unexplained and are routed for evidence-based review. Actual ceilings must come from the applicable AICTE handbook, approval letter and notifications.

9. APAAR Identity Resolution and Exception Handling

1. Verify submitted APAAR through an authorised adapter. In the prototype, the adapter is clearly labelled as simulated.
2. Perform a configurable format pre-check separately from registry verification.
3. Tokenise using keyed HMAC; protect secrets through KMS/HSM, version key IDs and plan key rotation. HMAC is pseudonymisation, not anonymisation.
4. Match protected tokens inside a controlled service boundary. Institutions do not receive other institutions' tokens or records.
5. If APAAR is missing, Fellegi-Sunter matching on approved attributes produces candidates only; a human resolves ambiguity.

Case	System response
Verification API unavailable	Queue bounded retries; keep status Pending; display service outage, never label as identity fraud.
Ambiguous identity	Manual evidence review; no automatic merge.
Student disputes a match	Open dispute; suppress adverse conclusion; preserve both records.
Corrected data submitted	Create superseding version and re-run relevant checks.
Category evidence missing	Keep eligibility pending; do not allocate a category seat.
APAAR not applicable or creation failed	Use explicit exception status and follow applicable policy; never fabricate a replacement ID.

10. Explainable Integrity Intelligence

Layer	Method	Role
1. Deterministic validation	Rules, constraints and referential checks.	Hard validation and policy outcome.
2. Statistical anomaly signals	Isolation Forest; median/MAD peer comparison; cohort and time-series comparison.	Risk signal and prioritisation for review.
3. Temporal graph analysis	NetworkX in prototype; graph database only if scale requires.	Relationship evidence across protected student, institution, programme, session, event, seat category and case.

10.1 Transparent Integrity Score

A 0–100 score prioritises review using visible weighted components such as seat utilisation, supernumerary share, missing-APAAR rate, correction rate and withdrawal pattern. Confidence and regulatory severity are shown separately; they are not collapsed into one opaque number.

10.2 Dashboard explanation

- Show the factors that contributed to the score.
- Show peer group and comparison period.
- Show the applicable rule/model version.
- Link each alert to underlying records and evidence.
- Allow reviewer to dismiss, request correction or escalate with a recorded reason.

Human decision safeguard
An anomaly is not proof of fraud. Shared phone numbers, addresses or bank accounts are not standalone evidence. No student or institution is automatically declared fraudulent by the model.

11. AICTE Central Monitoring Dashboard — Enhanced

Version 2.1 specifies the operational views needed by AICTE officers, rather than referring only to a generic dashboard.

11.1 Executive overview

Dashboard card	What it displays
Institution coverage	Onboarded institutions, reporting institutions and pending submissions.
Student-level coverage	Student-level records received and percentage of reporting scope covered.
Aggregate reconciliation	Reconciled totals, unresolved variances and aggregate-only submissions.
Intake utilisation	Approved intake, active eligible enrolment, category pools and remaining capacity.
Identity verification	Verified, provisional, pending, failed and exception records.
Review workload	Open cases by reason code, age, priority and assigned reviewer.
Risk distribution	Institution/programme risk indicators with explanations and trend.
Data quality	Missing fields, invalid dates, duplicate rows and correction rates.

11.2 Filters and drill-down

- Institution, state/region if authorised, programme, academic year and seat category.

- Submission date, reporting cut-off, status and source.
- Risk priority, reason code, assigned reviewer and unresolved duration.
- Drill from institution summary to programme, submission batch, reconciliation run and authorised record-level evidence.

11.3 Role-specific views

Role	Primary dashboard actions
Institution Data Operator	Upload records, view preflight errors, see aggregate variance, correct and resubmit, track submission status.
AICTE Administrator/Reviewer	Monitor institution compliance, inspect reconciliation variances, review cases, approve/reject/return for correction and generate reports.
Authorised Verifier	Verify only purpose-approved minimal claims and view verification status.
System Administrator	Manage roles, configured rules, system health, integration status and audit activity.

12. Enhanced System Architecture

The architecture retains the original React/TypeScript, FastAPI, PostgreSQL, Pandas, scikit-learn and NetworkX stack. Version 2.1 makes the reconciliation and operational workflow explicit.

Layer	Components and responsibilities
1. User & onboarding layer	Institution portal, AICTE reviewer dashboard, authorised verifier portal; institution registration and role provisioning.
2. Ingestion layer	CSV/Excel upload, REST APIs, ERP adapters; assisted column mapping; preflight validation.
3. Reconciliation layer — strengthened	Aggregate submission store, student-level event store, scope alignment, variance calculation and reconciliation case management.
4. Identity & lifecycle layer	APAAR adapter, HMAC token service, candidate matching, TEIE episode builder and transfer classification.
5. Compliance & intelligence layer	SeatGuard policy master/ledger, deterministic checks, anomaly signals, temporal graph and explainable score.
6. Review & trust layer	AICTE review queue, correction workflow, RecordTrust audit chain, credential/verification service.
7. Data & reporting layer	PostgreSQL source of truth, encrypted PII store, metrics/read models, dashboards and reports.

Architecture principle
PostgreSQL remains the source of truth. Dashboards use derived read models/metrics. Stateless ingestion workers can scale independently. Redis, Kafka, Kubernetes or a graph database are added only when justified by measured needs.

12.1 Text architecture flow

Institution / AICTE / Verifier → Portal & API Gateway → Smart Ingest → Aggregate Reconciliation + APAAR Identity Resolver → TEIE → SeatGuard → Integrity Intelligence → Review Queue / RecordTrust / ConsentPass → Central Registry & AICTE Dashboard.

13. Data Model and API Contracts

13.1 Core entities

- Institution, Programme, Approved_Intake, Seat_Policy_Snapshot.
- Person (pseudonymous token + key version), Person_PII (separately encrypted).
- Submission_Batch, Aggregate_Submission, Reconciliation_Run, Reconciliation_Variance, Reconciliation_Resolution.
- Enrolment_Episode, Episode_Version, Enrolment_Event (effective_at, recorded_at, actor, evidence_ref, prev_hash).
- Validation_Finding, Match_Candidate, Review_Case, Institution_Metrics.
- Consent_Record, Credential, Audit_Log (previous hash, entry hash).

13.2 Prototype API contracts

Endpoint (illustrative)	Purpose
POST /api/v1/institutions/onboarding	Create/request institution onboarding.
POST /api/v1/submissions/preflight	Validate file/schema before submission.
POST /api/v1/submissions	Create student-level submission batch.
POST /api/v1/aggregates	Submit aggregate enrolment report.
POST /api/v1/reconciliation/run	Run aggregate-to-student-level reconciliation.
GET /api/v1/reconciliation/{id}	View reconciliation result and variance.
GET /api/v1/institutions/{id}/ledger	View institution seat ledger.
GET /api/v1/students/{id}/timeline	Restricted student enrolment timeline.
GET /api/v1/reviews	List authorised review cases.
POST /api/v1/reviews/{id}/decision	Record reviewer decision.
POST /api/v1/students/{id}/corrections	Create superseding correction.
POST /api/v1/verifications/verify	Verify purpose-approved credential claim.
GET /api/v1/audit/verify	Verify audit hash chain.

These are proposed prototype contracts, not existing AICTE endpoints.

14. End-to-End Workflow

1. Institution operator signs in and selects institution, programme, academic year and reporting scope.
2. Operator uploads student-level data, aggregate totals, or both.
3. Smart Ingest maps columns and runs preflight validation; invalid rows return reason codes.
4. Aggregate Reconciliation aligns scope and compares reported totals with eligible student-level counts.
5. APAAR Identity Resolver performs format checks and authorised/mock verification; unresolved identities remain explicit exceptions.
6. TEIE builds or updates enrolment episodes and classifies transfers, overlaps and duplicates.
7. SeatGuard checks permitted capacity using the correct policy snapshot and category pool.
8. Integrity Intelligence generates explainable risk signals and prioritises review.
9. Clean records are committed; variances and conflicts enter the relevant institution correction queue or AICTE review queue.

10 Reviewer approves, rejects or requests correction. Corrections create new versions; nothing is silently overwritten.

11 RecordTrust stores the audit event and dashboards refresh.

12 Where authorised and implemented, ConsentPass issues or verifies minimal-claim credentials.

15. Prototype Scope and Implementation Priorities

The enhanced proposal uses a strict build order. The team should demonstrate a complete working core loop before attempting every advanced component.

Tier 1 — Must build (working core loop)

- Institution and AICTE role login; basic institution onboarding/context selection.
- CSV upload with assisted column mapping and preflight validation.
- Aggregate report submission and aggregate-to-student-level reconciliation preview.
- TEIE episode builder with sequential transfer versus unexplained overlap classification.
- SeatGuard ledger with configurable category rules and policy snapshot trace.
- AICTE review queue, correction/resubmission workflow and one useful dashboard.
- Synthetic demo dataset with clear, labelled scenarios.

Tier 2 — Advanced prototype (if time allows)

- HMAC tokenisation with mock APAAR adapter and exception statuses.
- Fellegi-Sunter candidate matching for missing APAAR.
- Isolation Forest plus median/MAD peer comparison and transparent score.
- Hash-chained audit log with verify button.
- Versioned corrections and reconciliation resolution history.
- Concurrent last-seat demonstration using row locking and idempotency keys.
- Institution submission tracking and role-specific dashboard filters.

Tier 3 — Roadmap

- Full W3C Verifiable Credential issuance and student-facing consent/sharing portal.
- Live APAAR/ABC/DigiLocker integration after authorised access and conformance testing.
- ERP/university connectors and automated scheduled submissions.
- Offline preflight validator, change-point detection and natural-language queries.
- Production-grade scale-out, disaster recovery and multi-authority governance.

Scope discipline
Do not present Tier 2 or Tier 3 capabilities as already implemented unless the team has built and tested them. The presentation must clearly distinguish working prototype, simulated integration and future roadmap.

16. Demo Scenarios and Seven-Minute Script

Case	Scenario	Expected result
S1	Valid student record	Accepted after applicable checks.
S2	APAAR format invalid	Format error; separate from registry verification.
S3	Same student repeated in one file	Duplicate submission/idempotency finding.
S4	Withdrawal at College A, then admission at B	Sequential transfer when evidence and timeline support it.
S5	Prolonged overlapping active enrolments	Human review case.
S6	Foreign national without APAAR	Configured exception processing; no fabricated APAAR.
S7	Similar name/DOB, APAAR missing	Probabilistic candidate only; human resolution.
S8	Admissions above configured capacity	SeatGuard alert with policy and calculation evidence.
S9	Aggregate total 500, student-level count 472	Variance 28; request details/explanation; do not invent records.
S10	APAAR service unavailable	Pending status, bounded retry and outage message.
S11	Two requests compete for last seat	Only one reservation/commit succeeds in transactional demo.
S12	Unusual institution distribution	Explainable anomaly signal; reviewer decides.

Suggested seven-minute demonstration

Time	Action
0:00–0:40	Problem and central idea: identity match is not automatically duplicate enrolment.
0:40–1:25	Institution login, programme/year selection and messy CSV upload.
1:25–2:05	Column mapping, preflight errors and correction.
2:05–2:50	Aggregate report versus student-level count; show reconciliation variance.
2:50–3:50	Transfer versus unexplained overlap using TEIE.
3:50–4:35	SeatGuard category ledger and over-cap alert.
4:35–5:20	Explainable risk signal and AICTE review case.
5:20–6:10	Reviewer decision, correction version and audit history.
6:10–7:00	Dashboard summary, integration honesty statement and final impact.

17. Example Dashboard Content

Section	Example display (synthetic demo values)
Institution reporting	6 demo institutions; 5 submitted; 1 pending.
Student-level coverage	472 student-level active records received for one illustrative reporting scope.
Aggregate reconciliation	Reported aggregate 500; student-level eligible 472; unresolved variance 28.
Seat utilisation	Approved intake 100; admitted 120; configured approved supernumerary 10; 10 need explanation.
Identity status	Verified / Pending / Exception / Failed counts, with clear definitions.
Review queue	Unexplained overlaps, seat-capacity alerts, identity disputes and reconciliation variances.
Data quality	Missing fields, duplicate rows, correction rate and unresolved submission errors.

All sample values are illustrative synthetic data, not real institutional statistics or measured model performance.

18. Security, Privacy and Governance

Control	Implementation proposal
Authentication	OIDC/OAuth2 (Keycloak or equivalent); MFA for privileged roles; signed requests or mTLS for connectors.
Authorisation	RBAC plus institution-scoped ABAC; reviewers see assigned cases; verifiers receive purpose-approved claims only.
Encryption	TLS in transit, encryption at rest, field-level encryption for high-risk attributes; secrets in KMS/HSM.
Data minimisation	Do not collect Aadhaar, biometrics, phone numbers or addresses for fraud detection.
Audit	Actor, event, timestamp, source, rule/model version, evidence reference and before/after state.
Resilience	Backups, recovery drills, key backup, monitoring and incident response.
Privacy law	Design with DPDP Act 2023 and applicable DPDP Rules/commencement provisions; AICTE and counsel must confirm lawful basis, retention and child-data requirements.

Mandatory regulatory processing and optional student-initiated sharing are separate purposes and require separately established legal bases. Consent is not assumed to be the legal basis for every regulatory data-processing activity.

19. Verifiable Credentials and RecordTrust

- Issue a signed W3C Verifiable Credential enrolment attestation only after validation and required adjudication.
- Include only necessary claims such as issuer, subject reference, institution, programme, period, status, expiry and status information.
- Use purpose-limited disclosure: an employer sees only permitted enrolment claims; a regulator sees authorised category/evidence results where required.
- Public QR codes never contain raw APAAR or personal data.
- Use append-only hash-chained audit records with chain verification and periodic Merkle roots.
- A hash chain is tamper-evident, not tamper-proof. Do not place personal data on a public blockchain.
- A valid credential signature proves issuer and integrity, not AICTE regulatory compliance.

20. Evaluation Metrics

Objective	Metrics	Evaluation approach
Identity matching	Precision, recall, F1, false and missed match rates.	Synthetic ground truth; separate exact and candidate matches.
Conflict detection	Precision, recall, F1 and resolution time.	Transfer, withdrawal, overlap and legitimate parallel-study scenarios.
Aggregate reconciliation	Variance correctness, scope-mismatch detection and resolution time.	Hand-calculated aggregate/student-level cases and seeded incomplete records.
Seat compliance	Rule-outcome correctness and false over-cap flags.	Hand-calculated cases tied to policy snapshots.
Anomaly detection	Precision@k, recall@k, PR-AUC and false-positive rate.	Injected labelled cases compared with rule-only baseline.

Objective	Metrics	Evaluation approach
Reviewer trust	Reviewer overturn rate and case resolution time.	Track dismissed flags and reasons by institution type.
Platform	Upload latency, throughput, p95 response and concurrency correctness.	Load tests and last-seat race test.
Audit integrity	Tamper detection rate.	Modify a test event; chain verification must fail.

Evidence standard
Report only measured results. Synthetic data demonstrates method and functional behaviour; it does not prove real-world fraud-detection accuracy.

21. Rollout Plan

1. **Discovery:** confirm field dictionary, institution/programme identifiers, approved intake sources, rule sources, roles, reporting scopes, retention and authorised API access.
2. **Synthetic-data prototype:** implement the Tier 1 core loop and demo cases, including aggregate reconciliation.
3. **Shadow-mode pilot:** run with volunteer institutions in advisory mode only; no adverse action; measure variance resolution and reviewer overturn rate.
4. **Authorised APAAR/ABC integration:** after access approval, security assessment and conformance testing.
5. **ERP and university connectors:** onboard systems incrementally with schema mapping and data-quality monitoring.
6. **National deployment:** monitoring, support, analytics, operational controls and periodic policy updates.

22. Risks, Limitations and Mitigations

Risk	Mitigation
No production APAAR or AICTE approval feed	Labelled mock adapters; seek official approval before pilot/production.
Aggregate records cannot identify individual students	Preserve aggregate-only evidence; track variance; request detail; never fabricate identities.
Policy changes or misconfigured rules	Versioned effective-dated configuration with approval provenance; rules sourced from AICTE.
False-positive conflict flags harm legitimate mobility	Temporal rules, study-mode matrix, evidence, human adjudication and shadow-mode pilot.
Model bias or poor labels	Rule-first design, transparent baselines, stratified evaluation and no automated adverse decisions.
Token correlation or privacy breach	KMS/HSM, least privilege, purpose limits, encryption, logging, retention and incident response.
Late or malformed uploads mistaken for misconduct	Preflight, correction windows and separate data-quality metrics from compliance.
Legacy ERP adoption	Onboarding support, adapters, assisted mapping and staged rollout.
Prototype scope too large	Tiered delivery; demonstrate the core loop before advanced features.

The platform checks consistency and authenticity signals; it cannot prove the truth of every plausible submitted entry. Human review and audit trails address that limitation.

23. Research Grounding and Proposed Novelty

SATYA-APAAR builds on established work in academic credential systems, privacy-preserving record linkage, graph-based anomaly analysis, Isolation Forest and selective disclosure. The novelty claim is not that these individual technologies are new; it is the proposed combination and operational use for AICTE enrolment assurance.

Proposed contributions

1. **N1 — Policy-versioned seat integrity:** capacity is derived from approval evidence and effective-dated rules, not institution-submitted totals.
2. **N2 — Privacy-preserving temporal identity resolution:** separates identity matching from genuinely conflicting enrolments.
3. **N3 — Time-bounded EnrolmentGraph classification:** new admission, sequential transfer, transition overlap, policy exception, unexplained overlap and duplicate.
4. **N4 — Lifecycle-aware seat reconciliation:** seats change only through verified status events.
5. **N5 — Explainable evidence-linked risk review:** transparent peer-based scoring with a human decision on every case.
6. **N6 — Purpose-limited credentials:** claims are issued only after required validation and adjudication.
7. **N7 — Aggregate-to-student-level reconciliation:** preserves the distinction between reported totals and identifiable student records, tracks variance and supports staged migration from aggregate-only reporting.

Defensible novelty statement
SATYA-APAAR 2.1 moves AICTE from collecting enrolment records to reconciling them. It combines aggregate-to-student-level reconciliation, privacy-preserving APAAR identity resolution, time-aware enrolment history and policy-versioned seat capacity to distinguish legitimate student mobility from unexplained conflicts and excess admissions. Every exception carries evidence and is resolved through an accountable human-review process.

24. Final Positioning

SATYA-APAAR 2.1 is a continuously reconciled, privacy-preserving and auditable enrolment assurance platform for AICTE-approved institutions. It does not merely collect student data. It establishes traceability across institution submissions, student identities, admission episodes, approved seat capacity, aggregate reports and review decisions.

Final message
One student. One academic identity. Every admission traceable. Every aggregate reconciled. Every approved seat accounted for.

What the prototype must prove

- A messy institutional file can be mapped, validated and corrected.
- Aggregate totals can be reconciled against student-level records without inventing missing identities.
- A legitimate transfer is not automatically flagged as a duplicate.
- An unexplained overlap reaches a human reviewer with evidence.
- Seat capacity is checked against a traceable policy snapshot.

- AI risk signals are explainable and do not make final adverse decisions.
- A reviewer decision and subsequent correction are auditable.

What not to claim

- Do not claim live APAAR verification without authorised integration.
- Do not claim aggregate-only records have been converted into verified student identities.
- Do not claim HMAC is anonymisation.
- Do not claim a valid Verifiable Credential proves AICTE compliance.
- Do not claim synthetic test results prove real-world fraud-detection accuracy.
- Do not claim pre-admission prevention without integration into an authorised admission/reservation workflow.